

Designing Last-Mile Delivery Robot Systems: Teleoperation Interfaces, eHMI, and Operator-Bystander Interaction

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Abstract

As e-commerce expands, autonomous Last-Mile Delivery Robots (LMDRs) offer promising solutions for efficient, sustainable urban delivery. When encountering unexpected obstacles or technical malfunctions, remote operators (ROs) must intervene through teleoperation systems. During such interventions, ROs often need to communicate with bystanders, yet current methods prove inadequate in noisy urban settings, when language barriers exist, or when nuanced communication is required. This dissertation investigates LMDR design and teleoperation through three interconnected research directions: (1) LMDR teleoperation, (2) external Human-Machine Interface (eHMI) design for LMDRs, and (3) operator-bystander mediated communication through eHMI. These directions address critical gaps in understanding how LMDRs can operate safely and effectively in shared urban environments. This research aims to deliver evidence-based design guidelines for teleoperation interfaces and eHMI systems, with actionable recommendations supporting sustainable LMDR integration into urban environments.

CCS Concepts

• **Human-centered computing** → *Ubiquitous and mobile computing design and evaluation methods*; **Empirical studies in ubiquitous and mobile computing**; *HCI design and evaluation methods*; Graphical user interfaces; **Empirical studies in HCI**.

Keywords

Human-Robot Interaction, Interface Design, Teleoperation, external Human Machine Interface, Last Mile Delivery Robots, Operator-Bystander Communication.

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1 Introduction

As e-commerce expands, efficient and sustainable last-mile delivery has become a pressing challenge. last-mile delivery, the final and

often most expensive delivery stage, faces issues such as driver shortages and urban congestion [17]. Electrically powered last-mile delivery robots (LMDRs) offer a promising alternative (Figure 1) [1–3, 7, 9, 10] and have seen growing adoption, particularly since the COVID-19 pandemic [18]. However, there is broad agreement that some situations will continue to require human remote operators (ROs), due to the need for human judgment, communication, and ethical or legal decision-making [8, 11, 13].



Figure 1: Postmates (Serve robotics) Delivery Robot (Reprinted with permission from 10.Robot.Postmates.WDC.25October2017. 2017, Elvert Barnes via <https://flic.kr/p/CGiBYu>, accessed on 30 April 2025).

The proposed research explores three foundational domains: LMDR operation in urban environments, teleoperation in robotics, and external Human-Machine Interface (eHMI) design. This research aims to explore the overlapping regions where these domains converge, creating increasingly complex and understudied phenomena.

1.1 Intersection 1: Teleoperation of LMDRs

The first intersection examines challenges of teleoperating LMDRs in unpredictable urban environments.



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1.2 Intersection 2: eHMI of LMDRs

The second intersection focuses on eHMI design for LMDRs and how robot intentions are communicated to pedestrians and other road users in urban spaces.

1.3 Intersection 3: Teleoperation of LMDRs Using eHMI

This intersection examines situations in which ROs communicate with bystanders through the LMDR's eHMI during teleoperation. While teleoperation and eHMI design for LMDRs have each been studied independently, their combination introduces distinct and underexplored challenges. This interaction represents a unique form of human-to-human communication mediated by a robot using eHMI.

2 Research Questions

This research examines how LMDRs can operate safely, efficiently, and acceptably in shared urban environments where both autonomous behavior and human intervention are required. It focuses on the design of teleoperation and external communication mechanisms that support such hybrid operation. Specifically, the study investigates how teleoperation interfaces can enable ROs to intervene effectively, how robots can communicate their intentions to passersby in dynamic urban settings, and how eHMIs can be used to mediate this communication in ways that are understandable, socially appropriate, and supportive of safe navigation.

The overarching goal is to facilitate the integration of LMDRs into everyday urban life by supporting smooth interaction between robots, ROs, and pedestrians. This work aims to generate actionable insights that inform evidence-based design guidelines for LMDR teleoperation interfaces and eHMIs in urban contexts.

3 Work to Date

The initial work established a solid basis for the design of teleoperation UI as well as the initial comprehension of the LMDR eHMI design space.

3.1 Study 1: Stakeholder Analysis of LMDR Teleoperation

First, we investigated operational challenges faced by LMDR ROs in real-world deployments. In order to do so, we conducted 15 in-depth interviews with industry professionals, representing various stakeholders within the LMDR ecosystem. A thorough thematic analysis revealed four distinct operational modes: (1) Monitoring - Passive observation of autonomous operations; (2) Tele-assistance - High-level commands (e.g., "Stop", path drawing); (3) Tele-driving - Direct remote control of the robot using keyboard, mouse, and joystick; (4) Field Team Dispatch. This categorization informs the design of appropriate communication interfaces for each mode.

Additionally, we identified seven primary categories of scenarios requiring RO intervention: Interaction with People, Connectivity Issues, Blocked Routes, Road and Environmental Conditions, Complex Traffic, Rules and Regulations, and Robot Malfunctions.

A critical finding highly relevant to this dissertation was the documentation of scenarios requiring operator-bystander communication through the robot. The study found that some robots are equipped with microphones and speakers to enable communication with humans in their environment, in addition to various complex interaction scenarios where effective communication between operators and bystanders is critical for safe and efficient robot operation. This gap in understanding how to design communication systems for these triadic interactions underscores the need for research into eHMI systems that can facilitate effective operator-bystander communication.

3.2 Study 2: Participatory Design Workshops of Teleoperation Command Menus

Building directly upon the intervention scenarios identified, we aimed to: (1) identify key RO tasks for each intervention scenario, (2) map UI commands necessary to execute these tasks, and (3) understand users' mental models for organizing commands into intuitive menu structures that reduce cognitive load and enhance situational awareness. Hence, fifteen individuals (10 females, 5 males, age $M = 28.07$, $SD = 7.15$) participated in four participatory design workshops. Four workshops were conducted at the HCI Laboratory, University of Haifa. Following an introduction to LMDR teleoperation and the twelve intervention scenarios from Study 1, participants engaged in three collaborative activities: (1) Task Extraction - For each scenario, participants identified key tasks the RO would need to perform; (2) Command Mapping - For each task, groups mapped relevant UI commands; (3) Open Card Sorting - participants grouped commands into categories and assigned labels using open card sorting methodology.

We analyzed workshop artifacts (sticky notes, categorizations) and video recordings to identify: (1) tasks generated across groups, (2) commands identified, and (3) categorization patterns and consensus levels. Participants identified 35 distinct RO tasks across all intervention scenarios and 76 total teleoperation UI commands. Among the 28 commands used by all groups, only 5 received consistent categorization across all groups, indicating moderate consensus on command organization. The command set was refined and filtered and then analyzed using hierarchical clustering with Ward's method yielding eight primary command categories. The current results will be further validated and refined to determine optimal organization and labeling aligned with operators' mental models.

3.3 Study 3: Taxonomy for LMDR eHMI Design

The objective of this study was to develop a comprehensive taxonomy for LMDR eHMI design that systematically organizes the design space and supports the analysis, design, and evaluation of eHMIs for delivery robots in complex urban environments. The taxonomy was developed through: (1) systematic literature review of eHMI research (primarily focused on autonomous vehicles), and (2) synthesis with empirical findings from Studies 1, particularly the stakeholder interviews regarding human-robot-human interaction scenarios. We organized LMDR eHMI characteristics into a hierarchical structure spanning multiple dimensions. The taxonomy comprises seven core dimensions: Modality, Function/Purpose,

Target road users, Communication Mode, Communication Strategy, States/Status, and Placement. The taxonomy reveals that while considerable research attention has addressed visual modalities for conveying autonomous vehicle intent (particularly in road-crossing contexts), significantly less work has explored eHMI designs that facilitate operator-mediated communication with bystanders. The "Function/Purpose" dimension includes categories such as "request" and "social interaction" that directly map to scenarios identified in Studies 1 and 2 where operators need to communicate with the public. The taxonomy provides a structured framework that will guide the design and evaluation phases of this dissertation, enabling systematic exploration of how different combinations of modalities, functions, and communication strategies can support effective operator-bystander communication in various intervention scenarios.

Together, these studies establish a strong empirical foundation for the research.

4 Planned work

By using a User Centered Design methodology, we will use the accumulated knowledge from previous studies to design and evaluate a UI that will allow an efficient and user-friendly remote operation. For passersby, we will draw on existing eHMI literature and employ a workshop-based research methodology to design an eHMI tailored to LMDRs. The resulting eHMI designs will be evaluated as well. The following subsections describe the next phases of the research.

4.1 Teleoperation UI design and evaluation

This study aims to design and evaluate a teleoperation user interface (UI) for LMDRs that provides ROs with a functional and efficient work environment during intervention scenarios. We will define core UI components as well as command menu structure. The design process will synthesize findings from our preliminary research and literature review to create a high-fidelity interactive prototype. The evaluation will consist of usability testing with approximately 30-40 participants, conducted in laboratory conditions. Through scenario-based testing we will assess whether the proposed interface provides adequate usability and delivers necessary features and information for ROs to perceive the remote LMDR environment, understand intervention requirements, and effectively resolve remote scenarios. Three LMDR intervention scenarios will be used based on the preliminary research described in study 1: (1) A pedestrian blocks the road (with no legal option for the LMDR to bypass autonomously). (2) A static object blocks the road (with no legal option for the LMDR to bypass autonomously). (3) The recipient cannot take delivery because the cabinet door is stuck.

Two of these scenarios (1 and 3) involve interactions with passersby, where eHMI will be used as a potential communication solution. For each intervention scenario, the system will enable the participant to execute high-level commands via a command menu designed according to the findings of the preliminary research described in study 2. To support an interactive response to the participant's selections, pre-recorded videos will be displayed in the robot's camera input window, depicting the LMDR's actions following each

selected command. These videos will be filmed in advance in real urban environments and will cover all possible participant choices.

After a short familiarization phase participants will be asked to solve the three scenarios of the experiment. During the experiment, we will record task success rate, number of commands issued (compared to minimal commands required), and error count by error type. Upon completion of the tasks, users will be asked to fill out the following questionnaires: (1) NASA Task Load Index (NASA-TLX) [6] to measure cognitive load, (2) Situation Awareness Rating Technique (SART) [15], and (3) Human-Machine Interface Acceptance Questionnaire (HMIAQ) [16]. Finally, a semi-structured interview will be conducted.

Final conclusions will be drawn using a systematic qualitative (summative interview) and quantitative analysis (questionnaires and performance measurements). Quantitative data will be analyzed using descriptive and basic inferential statistical methods. Qualitative data will be analyzed to identify patterns that support or contradict the findings.

4.2 eHMI Design Hackathon

Prior to the eHMI evaluation phase, we will conduct a one-day design hackathon with design students. The hackathon will generate a range of eHMI design concepts through a structured design challenge for scenarios 1 and 3 from the teleoperation UI evaluation phase. The hackathon connects our taxonomy development (study 3) with the experimental evaluation. Rather than moving directly from theory to testing, we use this creative ideation phase to explore the design space before selecting specific prototypes for evaluation. The one-day hackathon will involve approximately 15-20 design students who will work in groups of 2-3 designers, leveraging collaborative design activity. Throughout the day, all design artifacts (sketches, storyboards, digital files, prototypes) are photographed, and all presentations will be video recorded for subsequent analysis. Analysis will identify recurring visual metaphors, interaction paradigms, and communication strategies across teams. Additionally, we will assess technical implementability of proposed designs given LMDR capabilities. Considering the winning designs and data analysis we will decide on one design of each scenario type which will be refined and translated into eHMI prototypes for empirical testing in the evaluation study. This participatory design approach ensures that our evaluation study tests designs that are not only theoretically grounded in our preliminary studies but also creatively developed through professional design and vetted by practitioners.

4.3 eHMI designs evaluation

Following the participatory design hackathon, this study aims to systematically evaluate the effectiveness of each of the eHMI designs providing empirical evidence about whether these design approaches effectively support communication in real-world intervention scenarios. Specifically, we will evaluate both eHMI designs independently to assess their performance across multiple dimensions: (1) bystander comprehension of robot intent, (2) appropriateness of intended bystander actions, (3) perceived clarity of the communication (7-point Likert scale), (4) willingness to cooperate with the robot (7-point Likert scale), (5) user experience

(UEQ-S), (6) trust in the robot-operator system (TiA), and (7) public acceptance (UTAUT2). Through controlled laboratory experiments with approximately 30–40 participants representing the general public, we will evaluate whether each design effectively facilitates operator-bystander communication during teleoperation scenarios. This study bridges design ideation and empirical validation, enabling evidence-based design recommendations. The findings will inform both the development of commercial LMDR eHMI systems and contribute theoretical understanding of (human)-robot-human communication in shared urban spaces.

Rather than implementing fully functional robots, we will use video-based simulation methodology, which has been validated in prior eHMI research [4, 5, 12] and allows for controlled experimental conditions while maintaining relative ecological validity [14]. Each prototype will be realized as a video sequence captured from a first-person pedestrian perspective in realistic urban settings. Videos will be presented on a large display screen, replicating the experience of encountering an LMDR on a sidewalk. This approach enables precise experimental control while approximating real-world conditions. This study will evaluate the same two intervention scenarios that were assigned to design teams in the hackathon:

Scenario 1: Blocked path: The LMDR stops due to a pedestrian blocking the sidewalk. The eHMI communicates a request to clear the path, and participants assess comprehension and likely response.

Scenario 2: Technical malfunction: The LMDR cannot complete a delivery due to a technical issue - a stuck door. The eHMI explains the situation, and participants assess comprehension and intended response.

By using identical scenarios across studies we can examine intervention situations from both sides of the operator-robot-bystander communication chain, providing comprehensive understanding of how design choices affect the entire interaction system. After a brief introductory video explaining LMDRs, participants will view both video recordings showing LMDR encounters from a first-person pedestrian or recipient perspective. For each video trial, participants will watch the complete video sequence, verbally indicate their understanding of what the robot is trying to communicate and describe what action they would take in response. Afterwards, they will rate the eHMI design based on the design's clarity on explaining the delivery robot's intent on a 7-Likert scale. Finally, participants will rate the subject's willingness to cooperate with the robot on a 7-Likert scale. After viewing all videos, participants will complete UEQ-S (User Experience Questionnaire), TiA (Trust in Automation), and UTAUT2 (Public acceptance) questionnaires.

Finally, a brief semi-structured interview will explore eHMI interpretation and attitudes toward delivery robots.

5 Discussion

This research aims to investigate LMDRs through three intersecting research directions that together converge to address the central question: how can LMDRs operate safely, efficiently, and acceptably in shared urban environments where both autonomous operation

and human intervention are necessary? The research delivers actionable recommendations for LMDR manufacturers, fleet management companies, and teleoperation service providers regarding: optimal teleoperation interface configurations, effective eHMI designs for both autonomous operation and operator-mediated scenarios, and strategies for ensuring public acceptance and trust. The multi-method approach combining industry stakeholder analysis, participatory design workshops, taxonomy development, and controlled empirical evaluation provides a replicable model for investigating complex socio-technical systems at the intersection of robotics, human-computer interaction, and urban operations. While focusing on LMDRs, the theoretical frameworks developed here extend to other teleoperated systems operating in human-populated environments. By investigating LMDRs through multiple intersecting research directions, this work establishes comprehensive foundations for understanding and designing effective autonomous delivery systems in shared urban spaces.

While this research is expected to make meaningful contributions, it has several limitations that define its scope and point to future work. The study focuses on two intervention scenarios and on LMDRs operating on sidewalks, which limits transferability of findings to other robotic platforms. The evaluation relies on laboratory-based, video-simulated scenarios, enabling controlled experimentation but not fully capturing the complexity of real urban environments or fully interactive communication. Cultural factors may also affect how eHMIs are interpreted in different contexts.

As LMDR technology continues to evolve, some intervention scenarios may be addressed through increased autonomy. However, situations requiring human intervention and communication with bystanders are likely to remain. This work does not address monitoring interfaces, operator training, or long-term public acceptance, which are important directions for future research.

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